Data m=0–3 fit = analytic tidal model (main.tex eq:tidal), term by term

Data (71 pts): $A = 0.9965 + 0.0296\cos\ell + 0.0434\sin\ell + 0.0636\cos 2\ell + 0.0366\sin 2\ell - 0.0299\cos 3\ell - 0.0114\sin 3\ell$, $R^2 = 0.915$

Model: $A/A_{\rm all} = N\,[\,1 + \alpha\kappa(3\mu^2-1) + \alpha\kappa\,\frac{d}{D}(\frac{9}{2}\mu - \frac{15}{2}\mu^3)\,]$, $\mu = \cos b_0\cos(\ell-\ell_0)$, $c \equiv \cos b_0$, $W_m = \frac{\sin(m\pi/4)}{m\pi/4}$

Projected on $\ell_0 = 15^\circ$ (data coefficient of $\cos m(\ell-\ell_0)$) = model expression = value at solution:

$m=0$:	0.9965	$= N[1+\alpha\kappa(\frac{3}{2}c^2-1)]$	$= 0.992\,[1+0.111(\frac{3}{2}\cdot 0.695-1)] = 0.9965$
$m=1$:	+0.0398	$= N\alpha\kappa\frac{d}{D}\,c\,(\frac{9}{2}-\frac{45}{8}c^2)\,W_1$	$= 0.992\cdot 0.111\cdot 0.812\cdot 0.834\,(4.5-5.625\cdot 0.695)\cdot 0.900 = +0.0398$
$m=2$:	+0.0733	$= N\alpha\kappa\,\frac{3}{2}c^2\,W_2$	$= 0.992\cdot 0.111\cdot 1.5\cdot 0.695\cdot 0.637 = +0.0733$
$m=3$:	−0.0292	$= -N\alpha\kappa\frac{d}{D}\,\frac{15}{8}c^3\,W_3$	$= -0.992\cdot 0.111\cdot 0.812\cdot 1.875\cdot 0.579\cdot 0.300 = -0.0292$

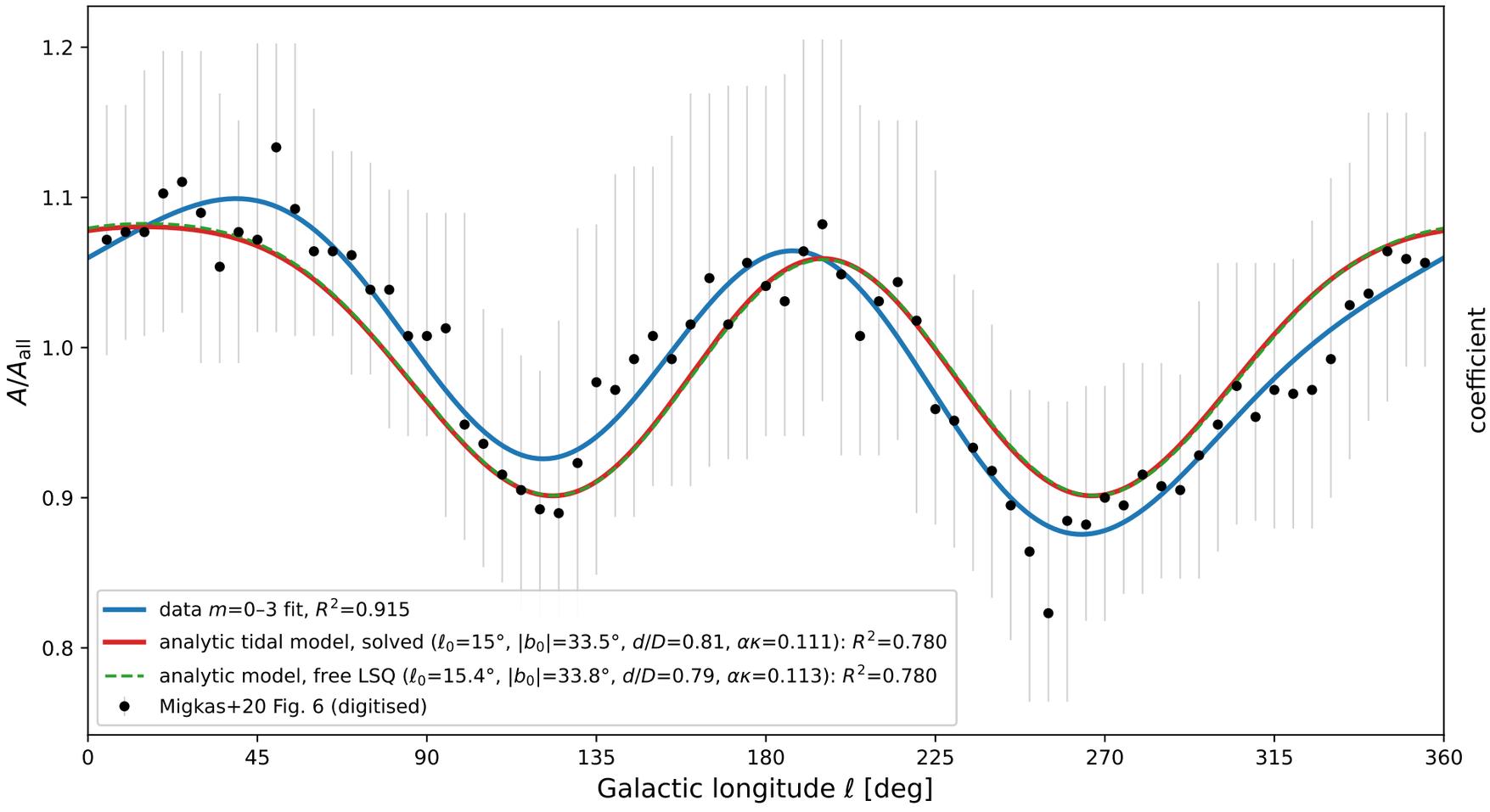
Solve: $m_1/m_3 \Rightarrow c^2 = 0.695$ ($|b_0| = 33.5^\circ$); $m_3/m_2 \Rightarrow d/D = 0.812$; $m_2 \Rightarrow N\alpha\kappa = 0.1105$; $m_0 \Rightarrow N = 0.9918$, $\alpha\kappa = 0.1114$

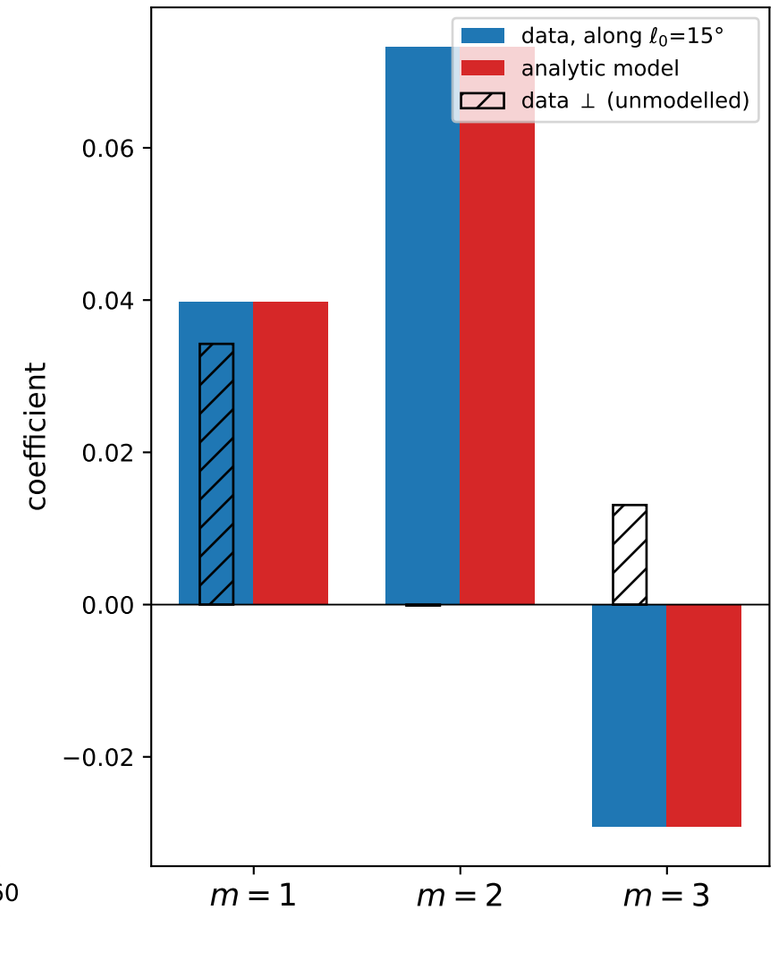
Not produced by an axisymmetric model (data $\sin m(\ell-15^\circ)$ parts): m_1: +0.0342, m_2: −0.0001, m_3: +0.0131

With the numeric code's 93.5° window ($W_{1,2,3}$=0.893, 0.612, 0.262): $c^2 = 0.706$ ($|b_0| = 32.8^\circ$), $d/D = 0.888$, $\alpha\kappa = 0.1143$.

Full numeric ideal-drag model, re-optimised (κ, d/D, D/L, ℓ_0) at fixed b_0: R^2 = 0.776 (b_0=0°), 0.777 (15°), 0.752 (33.5°, D/L at 14 bound)

→ b0≈33° does not help the numeric model (its m1 is already >0 along ℓ0 at b0=0, from higher orders); the truncated analytic form needs cos²b0<0.8 to get m1>0.





71 points come from overlapping 90° sectors (not independent). The analytic form is a small-d/D expansion evaluated at d/D≈0.8.